Paper 6.5

The Passive-Motion Fallacy

Kinetic Energy, the Ghost in the Reference Frame, and the Vindication of the Medium

Critique Companion to Paper 6 · Paper 6.5 of the Series

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ABSTRACT

Paper 6 established that there is no rest: rest and uniform motion are two active cycles of a ceaseless internal ledger, distinct conditions of the medium that conceal their difference from any internal observer because the directional offset is shared by everything in the frame. This companion paper turns that mechanical account onto the standard treatment of motion and finds, at the foundation, an unanswered question the standard model has never been able to answer: where, physically, does kinetic energy go when an object moves? Classical and modern physics treat kinetic energy as a scalar tacked onto an object by reference-frame bookkeeping — a number that appears with velocity, that can be made to vanish by choosing a frame in which the object is at rest, and that corresponds to no structural change in the matter itself, which is held to be identical at rest and in flight. The paper argues this is a conservation violation hiding in plain sight: a quantity that is real enough to do damage in a collision cannot also be a quantity that disappears when an observer changes frames, unless energy is not, in the standard account, a physically localized thing. The framework's law that energy is added geometry, conserved through relocation, forbids the evasion: kinetic energy is the physical directional offset written into a configuration's ledger, an absolute geometric fact that no change of frame can erase. From this the paper recovers and unifies what the standard account keeps separate — work, inertia, and the persistence of motion — as one conservation process, and exposes the passive-velocity picture as a delusion: a moving configuration is not coasting inertly but actively rewriting its identity across the medium on every tick, fueled by the geometry it absorbed when it was set in motion. The paper closes by entering into evidence four of the discipline's own giants — Mach, Lorentz, de Broglie, Dirac — each of whom fought the passive-empty-space picture and was overruled by the spacetime orthodoxy, and each of whom the framework vindicates by supplying the hardware their objections lacked. None of the audit requires alleging error in the predictions. It requires only asking the question the scalar conceals: when a thing moves, what physically changed?

1. The Question the Scalar Conceals

Standard physics has a formula for the energy of motion and no account of where it lives. A body at rest is assigned zero kinetic energy; a force does work upon it; the body acquires a velocity and, with it, a kinetic energy written as one-half its mass times the square of its speed. The formula is exact and the predictions it supports are not in dispute. What is in dispute is what the formula describes. The standard account treats the acquired kinetic energy as a scalar property of the system relative to a reference frame — a number associated with the object's velocity — and offers no physical description of where or how that energy is localized within the matter itself. The body is held to be structurally identical whether it sits still or travels at two thousand metres per second; velocity is a passive attribute layered onto unchanged matter, not an alteration of the matter.

This is a strange silence, and it is worth making the strangeness explicit, because familiarity has worn it smooth. The discipline that takes conservation of energy as one of its deepest laws cannot say, of the most ordinary energy there is — the energy of a moving thing — where that energy physically resides. It can compute the number with perfect accuracy and locate the number nowhere. The matter is the same; only the number has changed; and the number lives, on the standard account, in the relationship between the object and a chosen frame rather than in the object.

The framework's objection begins exactly here, and it is not an objection to the arithmetic. It is the observation that an energy located nowhere physical is an energy that has been removed from the conservation ledger and re-entered as bookkeeping — and that this is the same move, made at the most elementary possible scale, that the companion critiques of this series documented at every other scale: a quantity the account cannot localize is managed administratively rather than mechanically. Kinetic energy is the first administrative metric, hiding in the first mechanics lesson.

2. The Ghost Energy Paradox

The sharpest form of the objection is a contradiction internal to the standard account, requiring nothing from the framework to state.

Standard physics holds, as bedrock, that energy is conserved — it cannot be created or destroyed. Standard physics also holds, through Galilean and special relativity, that kinetic energy is frame-dependent — an object with a large kinetic energy in one frame has zero kinetic energy in the frame that moves with it. Place these two commitments side by side. A quantity that can be made to equal zero by a change of observer, and equal to a great deal by another change of observer, with nothing about the object itself altered between the two descriptions, is not behaving as a conserved physical substance. It is behaving as a number assigned by a perspective.

The standard account manages this with the doctrine that conservation holds *within* any single inertial frame, and that comparisons across frames are simply different valid bookkeepings. That is internally consistent, and it is exactly the tell. Precision is required here, because the point is easily mistaken for an error it is not. The framework does *not* claim that the frame-dependence of kinetic energy is an arithmetic inconsistency in relativity. Energy is the time-component of the energy-momentum four-vector; it is supposed to transform under a change of frame, vanishing in the object's rest frame, and that bookkeeping is perfectly self-consistent frame by frame. The framework grants this without reservation. The critique is not mathematical but ontological: that a quantity whose magnitude can be set to anything, including exactly zero, by an arbitrary choice of observer — with nothing whatever about the object altered — is being treated as a relational ledger entry rather than a located physical substance. The consistency of the bookkeeping is not the rebuttal to this point; it is the evidence for it. A theory that localizes its energies does not let them evaporate under a change of view. It means the conservation law is being applied to a quantity whose magnitude is set by the observer — which is to say the conservation is conservation of a bookkeeping entry, not of a localized physical thing. The energy of a moving object is real enough to shatter what it strikes, and the damage it does is not frame-independent in the standard telling either, but scales with the relative velocity of object and target. The account is consistent, but it has purchased its consistency by making the most ordinary energy in the world into something that has no fixed amount until an observer is chosen. The cost of the purchase is visible in any collision. A hundred-tonne train at speed carries energy enough to crush steel and generate heat; to a passenger riding with it, that same train carries exactly zero kinetic energy. When it strikes a wall and releases the energy as destruction, the standard account must say where that energy physically was the instant before impact — and on its own terms it was nowhere, a frame-relative scalar with no location in the train. The framework's objection is not that the number is wrong; the number is right. The objection is that a real capacity to do physical work cannot have been stored nowhere, and a theory that says it was stored nowhere has confused the ledger entry for the thing the entry counts.

This is also what answers the trap a critic builds with three observers rather than two: if A is stationary, B moves at one speed and C at another, must the medium hold several conflicting kinetic energies for one object at once? It holds none of them. The object carries exactly one absolute geometric offset, written once against the photon baseline. The several "observed" energies are not entries in the medium; they are the results of three different intersection calculations, each produced when a particular observer's partner-scan queries the object's. Each observer's own instrument acts as a local filter, and the filter's relative offset sets what that observer reads — but the underlying ledger entry is singular and absolute, and none of the readings is stored in the object. The multiplicity is in the comparisons, not in the medium; the medium keeps one book.

The framework refuses the purchase, because its law of conservation does not permit it. Energy is added geometry, conserved through relocation (Paper 4): a physical, localized displacement in the continuous medium, which can be transferred and rewritten but cannot be created, destroyed, or wished into a reference frame. Under that law, the kinetic energy of a moving configuration is the directional offset physically written into its ledger relative to the medium — an absolute geometric fact about how the configuration is querying its neighbors, present whether or not any observer is there to choose a frame. A second observer moving alongside cannot make the offset vanish; the observer can only fail to detect it, because the observer shares it (Paper 6, Section 4). The standard account confused the failure to detect with the absence of the thing. The offset is there; the medium carries it; the observer's inability to measure it from inside is not its non-existence but its self-concealment.

2.1 The Localization the Tensor Does Not Deliver

The strongest institutional reply must be met by name, or the critique looks naive. A physicist will answer that standard physics *does* localize energy: the stress-energy tensor assigns an energy density at every point, and its time-component is exactly the localized energy density the framework demands. On its face this closes the question.

A structural audit of the tensor for the case at issue reopens it. Evaluate that time-component for a moving, structureless point mass and it is the rest mass multiplied by a delta function — a zero-volume mathematical point — scaled by a frame-dependent factor. This does not exhibit an internal storage mechanism; it relocates the problem. A delta function is a spike of zero width, and packing a finite energy into zero volume is handled only by the formalism's tolerance for the unphysical; the construction places the energy "at a point" precisely because the point particle has no interior in which to place it. And because the scaling is frame-dependent, the energy density it reports rewrites itself across the whole assignment when the observer's motion changes — which means it carries the same frame-relativity the ghost-energy argument already identified, now wearing the prestige of tensor notation. If that density were a description of located substance, an observer could alter the energy density of a distant body by accelerating their own instruments, which no one believes. The tensor balances the bookkeeping frame by frame, exactly as the scalar did; it does not say what inside the particle is physically altered to sustain the energy. It formalizes the non-localization rather than curing it.

The framework's offset is the located thing the tensor gestures at and cannot supply: not a delta-function spike scaled by perspective, but a real directional asymmetry distributed across the configuration's pattern of partner queries, present whether or not an observer is there to choose a frame. Where the tensor places energy at a dimensionless point by mathematical fiat, the framework places it in the configuration's actual structure, where it can be conserved as the geometry it is.

2.2 A Consequence for the Baseline: The Shift Is in the Clock, Not the Photon

Establishing the photon as the sovereign zero invites one immediate objection from the gravitational side, and it is worth answering here because the answer follows the same logic as the kinetic-energy case. If light is the un-tilted baseline of the medium, why does a photon shift in frequency — red or blue — as it climbs out of or falls into a gravitational well? A baseline, the objection runs, should not change.

In the framework it does not. The photon's frequency is the medium's own baseline processing rate, and that rate is the standard against which everything else is measured; it is not itself a thing that drifts. What changes is the measuring clock. A clock or detector deep in a region of high mass density runs its own internal scan at an altered rate, because the surrounding loaded medium changes the local workload of its partner queries — exactly the mechanism Paper 10 develops for gravity. The photon arrives carrying the baseline it always carried; the apparatus reading it is running fast or slow relative to that baseline, and reports a shift. The observed red- or blue-shift is therefore a real, measurable effect — but it is a footprint of the clock's altered rate, not a change in the baseline itself. This is the same correction the framework makes for kinetic energy: the quantity standard physics attributes to the observed object (the photon's frequency, the train's energy) is relocated to where it physically belongs (the observer's local ledger rate, the configuration's own offset). The full gravitational treatment is the work of Paper 10 and is not pre-empted here; this notes only that the baseline survives the objection intact.

3. The Mechanical Transition: How Rest Becomes Motion

The standard account's silence is loudest at the moment of transition. It says a force does work and the object acquires velocity, but it does not say what physically happens inside the object as rest becomes motion — because, on its own terms, nothing does; the matter is unchanged and only the frame-relative number is different. The framework, holding to conservation of geometry, must give a mechanical account, and the account is the substance of this section.

A configuration in the active stationary state (Paper 6) runs a balanced cycle: its twin scans query their neighbors symmetrically, updates register evenly across the three axes, net position holds. For this configuration to move, three things must happen in sequence, and each is a step of strict conservation.

First, the injection of geometry. Motion cannot be self-started, because a balanced cycle has no surplus to spend; an external source — a photon, a colliding configuration — must inject actual physical geometry into the stationary configuration's boundary. This is the work the standard account names but does not localize: work is the literal delivery of geometry across the boundary, not the abstract product of a force and a distance.

Second, the absorption and forced integration. The configuration cannot hold the injected geometry as a passive passenger, set aside and carried inert, because it has no inert compartments — it must run its twin scans on every tick, and the new geometry is now part of what it scans. The injected displacement is forced into the processing cycle because there is nowhere else in a continuously scanning configuration for it to be.

Third, the transformation into an offset. To conserve the added geometry — to neither destroy it nor let it accumulate unaccounted — the configuration's ledger is permanently re-cut. The balanced cycle, whose updates summed to no net direction, is forced into an asymmetric cycle whose updates step along a vector. The injected geometry *becomes* the directional offset; it is not stored beside the motion, it is the motion. Kinetic energy is not added to a moving object as a label; it is the added geometry, now reading as the offset that steps the configuration across the medium tick after tick.

This also fixes what happens when two configurations carrying different offsets collide, a question the frame-relative account can answer only by switching bookkeepings. Here there is no master clock and no external referee — the continuous medium is the local referee. When two configurations collide they momentarily share a localized cluster of coordinates, and the local medium cannot support two conflicting update rates in the same coordinates at once. The re-balancing is a direct, local transfer: excess asymmetry in one ledger is forced into the available slots of the other, in full quantized steps, until the shared region runs a single consistent update again. Conservation is satisfied locally and mechanically, geometry relocated from one knot to the other, with no appeal to a global frame and no quantity that exists only relative to an observer. What prevents the configuration from simply fracturing under this forced integration is the Closure Principle (Paper 5): the configuration's boundary holds because the energy required to rupture its closure is greater than the kinetic geometry being delivered in the collision. The mismatch cannot break the boundary, so the ledger has no available move except to absorb the delivered displacement by cutting a new directional step into its running updates. Closure is what guarantees that a collision rewrites a cycle rather than shattering it — the configuration bends its motion because bending is cheaper than breaking.

This is why a moving configuration keeps moving with no continuing push, and the standard account's first law of motion falls out as a consequence rather than a postulate. The offset is conserved geometry; a conserved quantity in a self-sustaining cycle does not dissipate on its own; so the configuration continues to rewrite itself forward along the vector until a further interaction re-cuts the ledger again. Inertia of motion is not a tendency the object possesses; it is conservation refusing to spend what was not delivered.

4. The Passive-Velocity Delusion and the Unity of Work and Inertia

Two corrections to the standard picture follow immediately, and together they dismantle the passive picture of motion.

The first is the exposure of passive velocity as a delusion. The standard account treats a coasting object as doing nothing — inert, requiring no internal activity, simply possessing a velocity. The framework shows the opposite: a moving configuration is engaged in an intense, ceaseless computational act, actively reconstructing its entire identity from one neighboring shape of the medium to the next, on every tick, fueled entirely by the geometric offset it absorbed when it was set in motion. Coasting is not rest-with-a-number; it is the most active thing a configuration does short of being accelerated. The appearance of effortless gliding is the same illusion as the appearance of rest — balanced or offset, the books are turning over furiously beneath a placid surface. Nothing in this framework coasts, because nothing in this framework is ever passive.

The second is the unification of work and inertia, which the standard account isolates as separate laws — work as force through distance, inertia as the resistance in force-equals-mass-times-acceleration. The framework shows them to be two faces of one conservation event. When a configuration is accelerated, the work done on it is the physical act of carving new geometry into its ledger, and the inertial resistance felt during the push is the structural workload of that very carving — the cost of re-cutting an established balanced or offset cycle into a new one. They are not two laws that happen to involve the same mass; they are the delivery and the resistance of a single act of ledger-rewriting, necessarily equal because they are the same geometry counted from the two sides of the interaction. The standard account writes them as separate equations connected by the empirical fact that the same mass appears in both. The framework writes them as one process, and the shared mass is not a coincidence to be noted but the standing departure being re-cut, which is what mass is.

So acceleration, in this account, is not the passive assignment of a velocity vector to unchanged matter. It is the structural re-routing of a configuration's unceasing existence — the rewriting of what the configuration is perpetually doing, paid for in delivered geometry, resisted by the workload of the rewrite, and conserved absolutely from the first tick to the last.

4.1 The Processing Budget, Stated as a Conservation Constant

The mechanism by which a directional offset slows a configuration's internal clock can be anchored here without pre-empting the quantitative law, which is Paper 13's. The anchor is a conservation statement: a configuration has a fixed total processing budget per tick of the medium — a geometric constant, the same for every configuration because it is set by the medium's own rate (Paper 4). That budget is divided between two demands: the spatial-relocation workload of carrying the offset forward, and the temporal-verification workload of the internal identity scan. Because the total is conserved, the two trade against each other strictly: every increment of budget spent stepping the offset across the medium is an increment unavailable to the temporal scan, so a faster offset leaves a slower internal clock. This fixes the *form* of the relation — a budget split under a conserved total, the spatial and temporal workloads constrained to preserve one baseline identity check — and it is exactly the structure that yields a Lorentz-type factor when the division is computed. The computation, and the demonstration that the factor it yields is the measured one, is the work of Paper 13. What is fixed here is that the processing budget is not a metaphor but a conserved quantity with two competing claimants, so that time dilation is a bookkeeping necessity of a finite ledger, not a borrowed image.

5. The Dissent From Inside: Four Who Fought the Passive Void

The framework's quarrel with passive space and passive motion is not new; it is the resolution of a quarrel the discipline has carried, unresolved, for over a century. Four of its principal figures pressed exactly the objections this paper presses, and each was overruled not by evidence but by the ascendancy of the spacetime picture. The framework vindicates each by supplying the mechanism the objection lacked.

Mach, against absolute empty space. Mach rejected Newton's absolute space as an explanatory fiction: empty space, he argued, can do nothing to an object and cannot serve as the thing an object's motion is measured against, because one cannot measure against nothing. His positive proposal — that a body's inertia is determined by its relationship to all the other matter in the universe — was philosophical, lacking a hardware account of how the relationship could be enforced. The framework supplies it literally: there is no empty void, space is a dense continuous medium of shapes, and a configuration must query its neighbors on every tick simply to exist. Mach's relationship-not-property reading of inertia is, in this framework, the plain mechanical fact of the partner-scan architecture (Paper 6): inertia is a relationship with the surrounding fabric because identity itself is.

Lorentz, against motion as mere perspective. Lorentz held that uniform motion through the background medium produces a real, physical change in a moving object — that a moving clock runs slow because the forces holding it together are genuinely modified by its passage through the medium, not because perspective has shifted. The mathematics he built to describe this — the transformations that still carry his name — survived into relativity, but his physical interpretation was discarded in favor of a symmetric spacetime in which no frame carries a real offset. The framework vindicates Lorentz's interpretation: rest and uniform motion are genuinely distinct internal conditions, the moving configuration carries a real directional offset written into its ledger (Paper 6, Section 3), and the relativistic mathematics works precisely because that shared offset cancels from every internal measurement — not because motion is mere perspective, but because the real physical change conceals itself from observers who share it (Paper 6, Section 4). The math of relativity is kept; Lorentz's reading of what the math describes is restored.

De Broglie, against the inert particle. De Broglie held that matter is not inert and bullet-like but carries an internal frequency — a ceaseless oscillation tied to its mass — and spent his life seeking a "double solution" in which a particle is a localized, high-density feature of a continuous background field, perpetually oscillating. The orthodoxy kept his wavelength and discarded his ceaseless internal clock. This is not idle history: the internal clock carries experimental support, a channeling experiment with roughly eighty-MeV electrons finding a transmission resonance at the energy where the rate of atomic encounters matches the predicted internal-clock frequency — the relation now written as the Compton frequency, mass times c-squared over the reduced Planck constant. The framework is that double solution stated as hardware: a configuration is a localized feature of the continuous medium that never stops scanning, running its high-frequency partner query on every tick of the medium's rate (Paper 6, Section 2). The internal clock de Broglie posited is the manifold scan; the mass it is tied to is the standing departure the scan maintains.

Dirac, against the empty vacuum. Dirac's vacuum is not an empty stage but a fully populated sea, and his deeper lesson was that a particle can never be isolated from the background it sits in — it is in constant interaction with the vacuum, never alone. The framework takes the lesson to its conclusion: the configuration is not merely interacting with the medium, it is a feature *of* the medium, made of the same fabric as its surroundings and inseparable from them by construction. Dirac's already-noted objections to the administrative mathematics of the later theory (Papers 3.5, 4.5, 5.5) and his populated vacuum point the same way — toward a continuous, active medium that the passive-space picture cannot accommodate.

These four are entered as evidence of one thing: that the passive-void, passive-motion, inert-particle picture this paper rejects was rejected, in its time, by figures the discipline reveres, and was retained over their objections because the spacetime orthodoxy offered a mathematics that worked. The framework's contribution is not the objection, which is theirs; it is the mechanism that makes the objection constructive — a medium with an absolute baseline, configurations that exist only by scanning, and motion as a real geometric offset rather than a perspective. The giants saw what was wrong. The framework supplies what is right.

6. Conclusion: Motion Restored to the Ledger

The audit disputes none of the formulas. One-half mass times velocity squared computes the energy of motion correctly; the Lorentz transformations compute relativistic kinematics correctly; the predictions stand. What the audit disputes is the ontology the formulas have been wrapped in: that kinetic energy is a frame-relative scalar localized nowhere, that a moving object is unchanged matter carrying a passive number, that rest and motion are the same state seen from two angles, and that the relativity principle requires a warping spacetime to explain. Each of these is a way of managing motion administratively rather than mechanically, and each dissolves the moment energy is required to be a localized physical thing.

The framework requires exactly that. Energy is added geometry, conserved through relocation; kinetic energy is the directional offset carved into a configuration's ledger when geometry is delivered across its boundary; a moving thing is a thing perpetually rewriting its identity forward across the medium, fueled by what it absorbed; work and inertia are the delivery and the resistance of that single rewriting; and the relativity principle is the self-concealment of a real offset shared by every observer in the frame. Motion is returned from the reference frame to the ledger, where it can be conserved as the physical geometry it is. Mach, Lorentz, de Broglie, and Dirac each reached for this and lacked the hardware to hold it. The hardware is the continuous medium, and on it there is no passive motion, no frame-relative energy, and no rest — only the ledger, carrying its offsets absolutely, and never stopping.

References

This paper is the critique companion to Paper 6 and part of the *Departure From the Photon Baseline* series. It cites the companion papers by their published records, and the primary historical figures of Section 5.

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14. Dirac, P. A. M. (1930). "A theory of electrons and protons." *Proceedings of the Royal Society A*, 126(801), 360–365. (The populated vacuum / sea.)

Paper 6 establishes the partner-scan architecture, the two active states, the recovery of the relativity principle, and inertia as the workload of rewriting a cycle, all of which this paper draws on. Paper 4 establishes energy as added geometry conserved through relocation, the law on which the ghost-energy argument turns. The quantitative accounting of acceleration and momentum is the work of Paper 7; none is pre-empted here.

A. The Kinetic Energy Translation Ledger

The table maps the standard treatment of motion to its framework reading, in the manner of the ledgers of the earlier critique companions.

Standard treatment	Standard definition	Framework reading
Kinetic energy	Frame-relative scalar, one-half m v-squared	Physical directional offset carved into the configuration's ledger; absolute, not frame-relative
Rest (zero KE)	An achievable inert baseline state	The active stationary state: a balanced cycle, not an absence of activity (forbidden as true rest)
Work done on a body	Force integrated over distance	The literal delivery of geometry across the configuration's boundary
Inertia	Resistance to acceleration, F = ma	The structural workload of re-cutting an established cycle into a new one
Persistence of motion (first law)	A postulated tendency to continue	Conservation of the offset geometry in a self-sustaining cycle
Frame-dependence of KE	Different valid bookkeepings per frame	The shared offset self-concealing from internal measurement (Paper 6)
"Coasting"	Passive, activity-free gliding	Ceaseless active reconstruction of identity across the medium

B. The Ghost-Energy Contradiction, Stated as a Pair

This appendix states the central contradiction of Section 2 as two standard commitments that cannot both hold of a localized physical quantity, so the conflict can be read directly.

Standard commitment	What it asserts	What it requires of kinetic energy
Conservation of energy	Energy cannot be created or destroyed	That kinetic energy is a real, conserved physical quantity
Frame-dependence of kinetic energy	KE is zero in the object's rest frame, large in another	That the same object's kinetic energy has no fixed value until a frame is chosen

A localized physical substance cannot both have a fixed conserved amount and have no fixed amount until an observer is selected. The standard account holds both by restricting conservation to within-frame bookkeeping — which resolves the contradiction only by conceding the point: the conserved quantity is a frame-relative entry, not a localized thing. The framework removes the contradiction at its root by making kinetic energy the configuration's real directional offset (Paper 6): an absolute geometric fact, conserved across all descriptions, that a co-moving observer fails to detect but cannot abolish. The offset is there in both rows; only the observer's access to it changes.

C. The Four Allies and the Mechanism Each Lacked

This appendix sets the four historical objections beside the specific framework mechanism that completes each, so the vindication is exact rather than rhetorical.

Figure	The objection to the standard picture	The mechanism the framework supplies
Mach	Empty space cannot act on a body or serve as a reference; inertia must be a relationship to all other matter	The partner-scan architecture: a configuration exists only by querying its neighbors, so inertia is literally relational (Paper 6, Section 2)
Lorentz	Uniform motion through the medium is a real physical change, not a perspective	The asymmetric offset cycle: motion is a real directional offset in the ledger, concealed by sharing (Paper 6, Sections 3–4)
de Broglie	Matter carries a ceaseless internal frequency tied to its mass	The manifold scan: a high-frequency partner query on every tick of c, maintaining the standing departure that is mass (Paper 6, Section 2)
Dirac	The vacuum is populated; a particle cannot be isolated from the background	The continuous medium: a configuration is a feature of the fabric, not a tenant in a void, inseparable by construction (Paper 6, Section 2)

Each row is the same shape: a figure who saw that the passive-void picture was wrong, an objection that lacked a hardware mechanism in its own time, and the mechanism the framework supplies a century on. The pattern is the argument: the framework is not introducing a novel quarrel with the standard account of motion: it is resolving an old one in favor of the dissenters.

D. The Mathematical Audit of the Relational Vector

This appendix shows, in explicit coordinate terms, why an offset that is absolute in the medium reads as relational to every internal observer — and why the comparison reproduces the familiar transformation results without any appeal to a warping of space or time. The notation is kept deliberately elementary, because the point is structural, not computational.

Let the photon baseline define the absolute reference of the medium. Each configuration carries one absolute directional offset written against that baseline: call them v-sub-A, v-sub-B, and so on, each a real ledger entry, not a frame choice. These are the quantities the medium actually holds.

Now let observer A measure configuration B. Measurement, in this framework, is not the reading of a coordinate; it is a partner-scan query — A's scan comparing its own ledger against B's across their shared boundary. The quantity that the comparison returns is the *difference* of the two ledgers across that boundary, because a partner-scan registers only what differs between the querying configuration and the queried one. Write the returned reading as

v(A reads B) = v-sub-B – v-sub-A.

Every term A's instruments can access enters through this difference, because every instrument A owns is itself a configuration carrying v-sub-A, and every query it runs is a comparison against its own ledger. The absolute baseline term does not appear in any reading A can take, not because it is absent, but because it is common to both sides of every comparison A is structurally able to make. This is the cancellation stated as algebra: a quantity present identically on both sides of a difference is not in the difference. When the relative offsets are small against the medium's rate, the relation above is exactly the Galilean velocity-addition an observer expects; when they are not, the same partner-scan comparison — now accounting for the budget split of Appendix-adjacent Section 4.1, in which carrying an offset costs internal-scan rate — returns the Lorentz-type relation instead, because the clocks doing the comparing are themselves slowed by their own offsets. The transformation is not imposed on space and time; it falls out of comparing two real ledgers with instruments that each carry one of them.

Three consequences are worth stating. First, the absoluteness and the relativity are both real and not in tension: the offset is absolute in the medium and relational in every internal reading, exactly as the body of the paper claims, because reading is differencing. Second, no observer is privileged, which is the relativity principle — every observer carries an offset, every observer differences against their own, and none can see their own baseline term, so the laws read identically for all (Paper 6, Section 4). Third, the energy follows the same pattern as the velocity: the kinetic energy A attributes to B is computed from v(A reads B), the differenced quantity, which is why it varies by observer while the offset that physically loads B does not. The energy is absolute in the medium and relational in the reading, for the identical reason the velocity is.

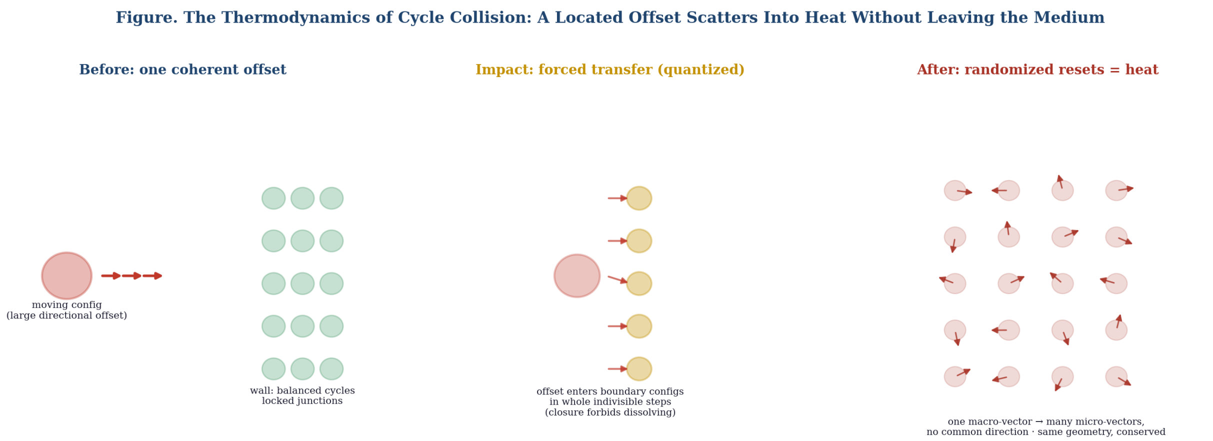

Figure E1. A located offset becoming heat without leaving the medium: the coherent directional offset of the moving configuration (left) transfers in whole quantized steps into the wall's boundary configurations on impact (centre), then cascades through the junctions into many small randomized multi-directional resets (right). One macroscopic vector becomes many microscopic ones; the geometry is conserved in full at every step. The randomized end state is what the framework means by heat.

This appendix details what happens, mechanically, when a moving configuration strikes a stationary one — the locomotive into the wall — so that the conversion of a located offset into heat and structural failure can be followed step by step, with the energy never once leaving the medium or losing its localization.

Begin with the two participants in the framework's terms. The moving configuration runs an asymmetric update cycle: its ledger carries a large directional offset, stepping it across the medium each tick (Paper 6). The wall is an assembly of configurations running balanced cycles, holding net position, bound to one another at locked junctions (Paper 5). The collision is the forced meeting of these two cycle types in a shared cluster of coordinates.

Step one: the mismatch. When the moving configuration's boundary enters the wall's coordinates, the local medium is asked to run two incompatible update rates in the same place at once — a stepped asymmetric cycle and a balanced one. It cannot. The Closure Principle forbids either configuration from simply dissolving, so the conflict cannot resolve by one party vanishing; the delivered offset must go somewhere, and the only somewhere available is the wall's ledgers.

Step two: the forced transfer. The directional offset is a quantized, indivisible entry (Paper 6, Section 3), so it cannot smear gradually into the wall; it transfers in whole steps into the boundary configurations it meets first. Those configurations, which had been running balanced cycles, are abruptly forced to absorb a directional step they did not carry — their books, balanced an instant before, now hold an offset they must account for.

Step three: the cascade. A boundary configuration that has absorbed more offset than its own closure can carry as a single coherent vector cannot hold it as clean directed motion. It sheds the excess the only way the medium allows: by transferring quantized steps onward to *its* neighbors, in whatever directions the local junctions present. What began as one large offset pointing one way is thus split, at each junction, into many smaller offsets pointing many ways, propagating outward through the assembly as a spreading front of ledger resets.

Step four: heat, defined. The end state of that cascade is an assembly whose configurations are now carrying many small, randomized, multi-directional offsets instead of one large shared one — high-frequency ledger resets with no common direction. That is precisely what the framework means by heat: not a fluid that flowed in, but the original directional offset, conserved in full, redistributed from one coherent macroscopic vector into a multitude of incoherent microscopic ones. The structural failure — the bending steel, the shattered junction — is the same event seen at the scale of the bonds: junctions whose delivered offset exceeded their closure threshold rupture (Appendix E's cascade is the closure threshold of Paper 5 being crossed locally), and the assembly reorganizes.

The accounting closes exactly. The geometry that was the moving configuration's offset before impact is, after impact, the sum of all the randomized offsets and broken-junction reorganizations — the same geometry, never created, never destroyed, never anywhere but in the medium. The destruction the standard account attributes to a frame-relative scalar is, in the framework, the located offset doing precisely what a conserved located quantity must do when it is forced to relocate all at once: it scatters, and scattering at the scale of bonds is heat and breakage. The energy that smashed the wall was inside the train the whole time, written in its ledger as the offset, exactly where the framework said kinetic energy lives.